

MP1994V2 Proof Ledger

Mortensen–Pissarides (1994) Lean replication

29 July 2026

Purpose. The milestone tracker records project state. This cumulative ledger records exactly what Lean has proved, the hypotheses used, and the economic meaning of each theorem.

1 M1: Surplus Bellman equation

Paper reference	Section 3, journal page 401, equation (8)
Adequacy status	COMPLETE – GREEN
Lean source	MP1994V2/SteadyState/Surplus.lean

1.1 Mathematical statement

For the actual equilibrium surplus

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U,$$

Equation (8) is a conditional representation theorem: for every object satisfying **ValueEquilibrium** **P**, Lean proves that the surplus derived from its value functions satisfies, for every idiosyncratic state ε ,

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda \int \max\{S(x), 0\} dF(x) - \beta\theta q(\theta)S(\varepsilon_u). \quad (8)$$

Here **P.shock** is the probability law of the idiosyncratic shock; its induced CDF is F , and integration with respect to dF is represented in Lean by integration against **P.shock**. Also, $\theta q(\theta)$ is **P.workerMeetingRate** **E.theta**. M1 does not prove existence of a **ValueEquilibrium**.

1.2 Economic interpretation

- $p + \sigma\varepsilon$: current match output.
- b : unemployment opportunity cost.
- $\lambda \int \max\{S(x), 0\} dF(x)$: option value of future idiosyncratic draws, allowing rejection of nonpositive continuation surplus.
- $\beta\theta q(\theta)S(\varepsilon_u)$: the employed worker’s foregone search gain at the upper-support job.

1.3 Exact Lean declarations

Role	Declaration
Continuation decomposition	MP1994V2.continuationTerm_eq_integral_activeSurplus_sub
Firm surplus share	MP1994V2.ValueEquilibrium.firm_share
Algebraic surplus flow	MP1994V2.ValueEquilibrium.surplus_flow_equation
Minimal probability theorem	MP1994V2.ValueEquilibrium.surplus_bellman_of_probability
Paper-level wrapper	MP1994V2.ValueEquilibrium.surplus_bellman

1.4 Assumptions

Used:

- equation (3), through the definition of actual surplus;
- equation (4), only at the upper state;
- equations (5) and (6), added to cancel the wage;
- equation (7), subtracted to account for unemployment value;
- probability normalization of `P.shock`;
- `ValueEquilibrium.active_surplus_integrable`.

The minimal theorem assumes only `[IsProbabilityMeasure P.shock]`. The wrapper takes `D : ShockAssumptions P`, installs `D.isProbability` locally, and uses no other field of `D`.

Not used: parameter signs; `CoreEconomicAssumptions`; upper-support restrictions or maximality; atomlessness; mean zero; unit variance; `ShockNormalizationAssumptions`; matching positivity or monotonicity; `MatchingAssumptions`; equations (1) and (2).

1.5 Readable proof

Write the shared continuation term as

$$C(\varepsilon) = \int [\max\{S(x), 0\} - S(\varepsilon)] dF(x).$$

Adding equations (5) and (6) cancels the wage and sums the Nash weights:

$$r[J(\varepsilon) + W(\varepsilon)] = p + \sigma\varepsilon + \lambda C(\varepsilon).$$

Subtracting equation (7), and using equation (3), gives

$$rS(\varepsilon) = p + \sigma\varepsilon - b + \lambda C(\varepsilon) - \theta q(\theta)[W(\varepsilon_u) - U].$$

Equation (4) at ε_u gives

$$W(\varepsilon_u) - U = \beta S(\varepsilon_u).$$

Probability normalization of `P.shock` and active-surplus integrability give

$$C(\varepsilon) = \int \max\{S(x), 0\} dF(x) - S(\varepsilon).$$

Substitution and moving $-\lambda S(\varepsilon)$ to the left proves (8).

1.6 Adequacy and next dependency

The actual `ValueEquilibrium` surplus is used. Equation (8) is derived, not added as an assumption or field. No cutoff, reservation rule, or affine surplus result is used. It is conditional on an existing `ValueEquilibrium` and does not establish that such an equilibrium exists.

Human-review conclusion. Equation (8) matches the paper and is derived non-circularly from equations (3)–(7). The actual equilibrium surplus is used, with no cutoff, affinity result, or reduced equilibrium. The theorem is explicitly conditional on an existing `ValueEquilibrium`.

M2 may subtract (8) at two productivity states to derive the surplus-difference identity, then prove affine surplus and monotonicity consequences. No M2 result is implemented here.

Review space:
